ANDONG HUA

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EDUCATION

University of California, Santa Barbara *Jan. 2024 – June. 2027 (expected)*

Ph.D. Candidate, Final Year | Trustworthy & Multimodal LLMs, AI for Healthcare | Advised by Prof.Qin Yao

University of California, Los Angeles *Sept. 2020 – Jun. 2022*

M.S. in Electrical and Computer Engineering, GPA: **3.95/4.0**

University of Nottingham *Sept. 2016 – Jun. 2020*

B.Eng. in Electrical and Electronic Engineering (2+2), GPA: **85/100**; Rank: **1/160 (Year 3)**

INDUSTRY EXPERIENCE

Amazon | Applied Scientist Intern | AWS Fundamental Research *June–Sept. 2026*

RL Credit Assignment for LLM Agent Post-training

- Demonstrated that PPO critics primarily act as turn-level credit assigners for LLM agents.
- Quantified turn-level credit assignment in LLM agent training using advantage similarity, and showed that higher similarity to ground-truth advantages strongly correlates with better policy performance.
- Identified complementary strengths of GRPO and PPO in advantage estimation, combining them via level–shape decomposition to achieve SOTA performance on WebShop and ScienceWorld.

Amazon | Applied Scientist Intern | Smart Vehicle Team *June–Sept. 2025*

Automated Prompt Optimization for LLMs

- Developed automated LLM prompt optimization methods using learned guidelines and error categorization, improving accuracy by 20% and supporting production use across multiple Amazon teams.
- Introduced an online prompt optimization framework for LLMs that achieved state-of-the-art results while reducing computational cost by over 10×.

TuSimple | Research Engineer (Full-time) | Deep Perception *July 2022–Feb. 2023*

- *Video Segmentation Co-training.* Developed and trained a multi-task deep learning model (U-Net) with a shared encoder and task-specific heads, using FP16 for efficient distributed training and inference.
- *Interactive Segmentation.* Built an interactive image segmentation tool to streamline data annotation, reducing the required clicks by up to 10×.

SELECTED PUBLICATIONS

Tang, K.*, **Hua, A.***, et al. SFT-as-Context Mitigates Forgetting in Supervised Fine-tuning. **ICLR 2027 submission**.

Hua, A., et al. TokenSwap: Benchmarking and Reducing the Modality Gap in Multimodal LLMs. **NeurIPS 2026**.

Hua, A.*, et al. Flaw or Artifact? Rethinking Prompt Sensitivity in Evaluating LLMs. **EMNLP 2025**.

Hua, A.*, et al. NutriBench: A Dataset for Evaluating Large Language Models on Nutrition Estimation from Meal Descriptions. **ICLR 2025**.

Hua, A., et al. Initialization Matters for Adversarial Transfer Learning. **CVPR 2024**.

SELECTED RESEARCH EXPERIENCE

University of California, Santa Barbara *Sept. 2026–Present*

LLM-based World Modeling for Personalized Glucose Forecasting

- Developing an LLM-based, action-conditioned world model for personalized glucose time-series forecasting, leveraging historical CGM data, planned interventions, and patient-specific trajectory retrieval.
- Building a large-scale forecasting benchmark from 5.8 million longitudinal records of 493 patients, integrating multimodal health data including glucose, meals, exercise, and insulin.

TECHNICAL SKILLS

Programming & Frameworks: Python, C/C++, PyTorch, TensorFlow, Hugging Face Transformers, Ray, verl, slime, vLLM, Scikit-learn, Distributed Training